

# INTRODUCTION TO BIOINFORMATICS

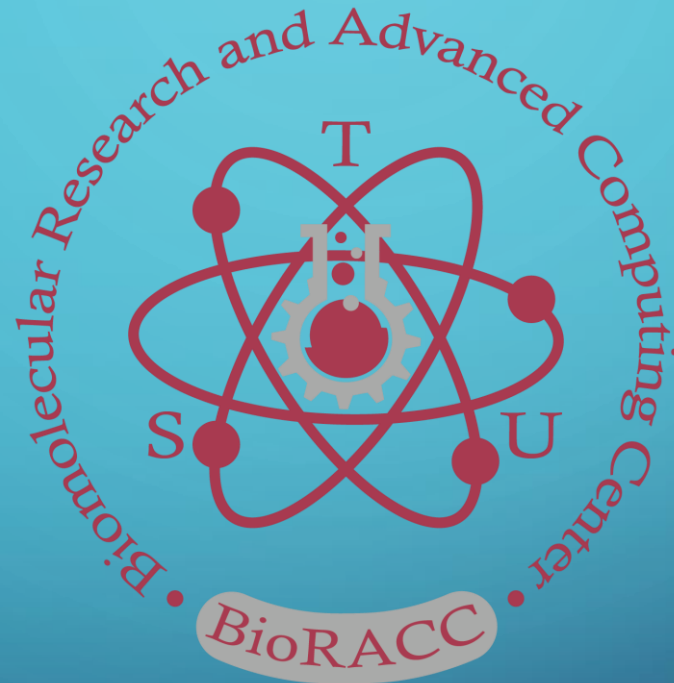
LINUX & HPC

# BIOINFORMATICS



- Intersection of biology, statistics/data science, and computer science
- “Omics”:
  - DNA-sequencing – mutations; epigenetics
  - RNA-sequencing – gene expression; isoforms
  - Metabolomics – metabolites/small molecules
  - Proteomics – proteins; PTMs
- Structural biology
- Systems biology

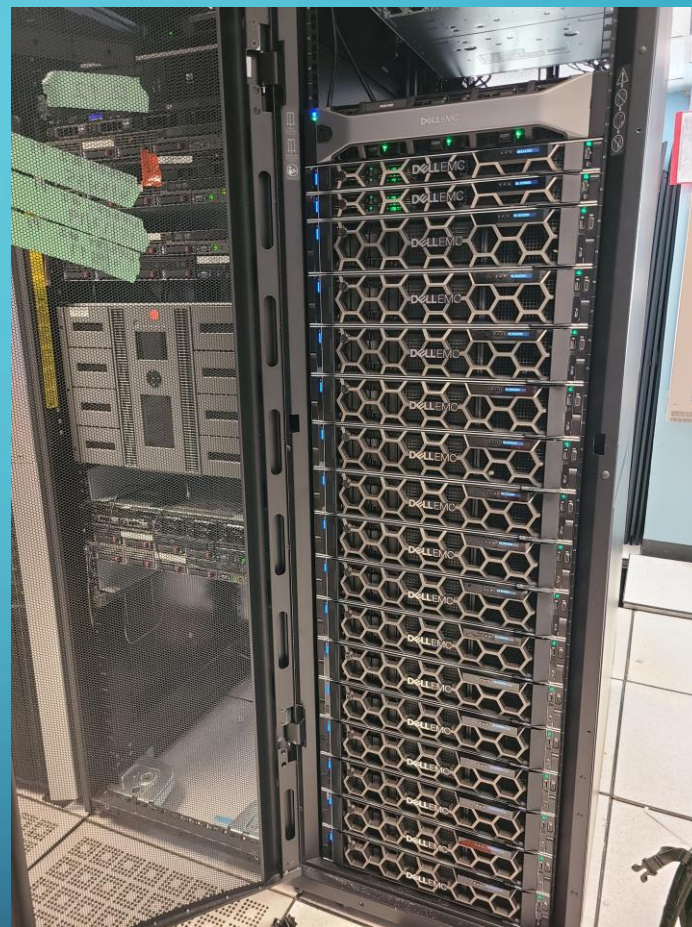
# BIORACC HIGH PERFORMANCE COMPUTING CLUSTER (HPC)



Solutions through the Lens of Science

# HIGH PERFORMANCE COMPUTING CLUSTER (HPC)

- 15 nodes:
  - 32-core Intel processor
  - NVIDIA A30 Graphics Processing Unit (GPU)
  - 256 GB RAM
- Shared 73 TB memory
- Slurm Workload Manager
- Open OnDemand web portal



# SECURE SHELL (SSH)

- Remotely access another computer/server/cluster
- macOS has a built-in SSH client called Terminal
- Windows has PowerShell
- Third party SSH clients have more features (e.g. MobaXterm or PuTTY)
- Public-key cryptography for a secure connection

• Computer 1



• Computer 2



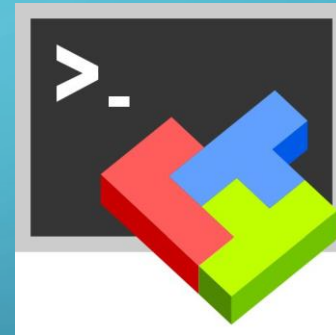
# ACCESSING REMOTE SERVERS/CLUSTERS WITH SSH



Mac  
Terminal



Windows  
PowerShell



MobaXterm



PuTTY

OPEN

**OnDemand**

OnDemand provides an integrated, single access point for all of your HPC resources.

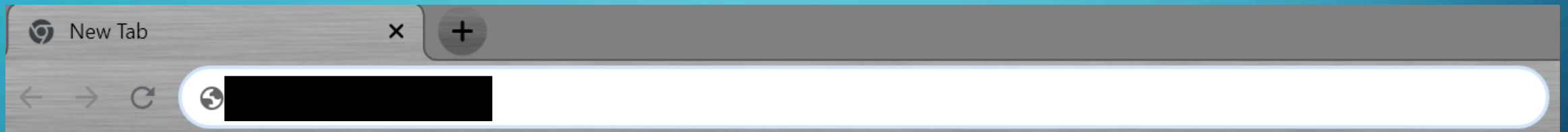
# ACCESSING THE CLUSTER VIA OPEN ONDEMAND

OPEN

**OnDemand**

OnDemand provides an integrated, single access point for all of your HPC resources.

1. Type `https://[REDACTED]` in a browser



Must be connected to TSU-NET secure network or the VPN to access cluster



# LOGGING IN

Username: firstname.lastname (all lowercase)



**Sign in to access this site**  
Authorization required by   
Username   
Password

# SUCCESSFULLY LOGGED IN



TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ 

 ▾  



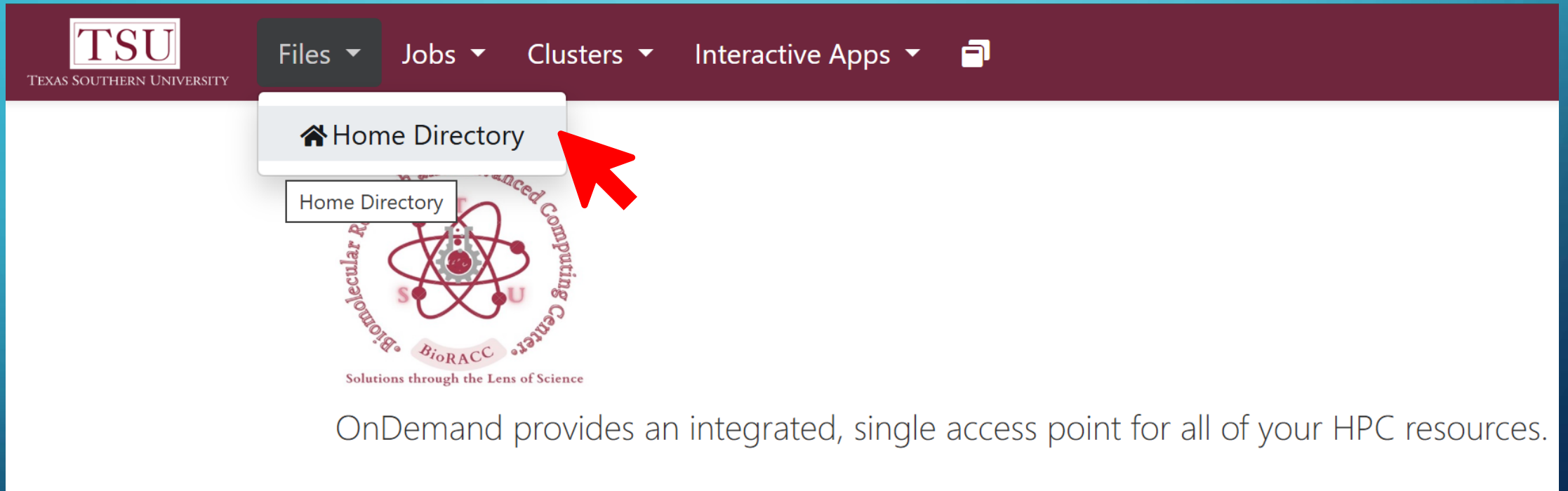
Solutions through the Lens of Science

OnDemand provides an integrated, single access point for all of your HPC resources.

powered by  
 

OnDemand version: 3.0.1

# HOME DIRECTORY



The screenshot displays the OnDemand web interface for Texas Southern University (TSU). The top navigation bar is dark red and contains the TSU logo, a 'Files' dropdown menu, and links for 'Jobs', 'Clusters', 'Interactive Apps', and a document icon. The 'Files' dropdown is open, showing a 'Home Directory' option with a house icon, which is highlighted by a red arrow. Below the navigation bar, the main content area features a large, faint circular logo for the 'BioRACC' (Biomolecular Research Advanced Computing Center) with the text 'Solutions through the Lens of Science'. Below this logo, the text reads: 'OnDemand provides an integrated, single access point for all of your HPC resources.'

TSU  
TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾

Home Directory

Home Directory

BioRACC  
Solutions through the Lens of Science

OnDemand provides an integrated, single access point for all of your HPC resources.

# CONTENTS OF YOUR HOME DIRECTORY

 TSU  
TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ 

 ▾  

 Open in Terminal ▾  Refresh  New File  New Directory  Upload  Download  Copy/Move  Delete

Home Directory

 / [home](#) / [databiom](#) /  Change directory  Copy path

☐ Show Owner/Mode ☐ Show Dotfiles Filter:

Showing 1 of 11 rows - 0 rows selected

|                          | Type                                                                                 | Name                                                                                                          | Size | Modified at           |
|--------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|------|-----------------------|
| <input type="checkbox"/> |  | <a href="#">ondemand</a>  | -    | 1/26/2025 12:00:22 PM |

powered by 

OnDemand version: 3.0.1

# ACTIVE JOBS

TSU

TEXAS SOUTHERN UNIVERSITY

Files ▾

Jobs ▾

⌚ Active Jobs

🔧 Job Controller

Clusters ▾

Interactive Apps ▾

 ▾





Active Jobs

➤ Open in Terminal

↻ Refresh

+ New File

➤ New Directory

⬆ Upload

⬇ Download

📄 Copy/Move

🗑 Delete

Home Directory

⬆

/ home / databiom /

📄 Change directory

📄 Copy path

☐ Show Owner/Mode

☐ Show Dotfiles

Filter:

Showing 1 of 11 rows - 0 rows selected

|                          | Type | Name     | Size | Modified at           |
|--------------------------|------|----------|------|-----------------------|
| <input type="checkbox"/> | 📁    | ondemand | -    | 1/26/2025 12:00:22 PM |

powered by

OPEN

 OnDemand

OnDemand version: 3.0.1

# SEE RUNNING OR QUEUED JOBS



Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ 

Your Jobs ▾ All Clusters ▾

Active Jobs

Show 

50 ▾

 entries 

Filter:

| ID                         | Name | User | Account | Time Used | Queue | Status | Cluster | Actions |
|----------------------------|------|------|---------|-----------|-------|--------|---------|---------|
| No data available in table |      |      |         |           |       |        |         |         |

Showing 0 to 0 of 0 entries 

Previous Next

powered by 

OnDemand version: 3.0.1

# JOB COMPOSER

**TSU**  
TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ 

 ▾  

Active Jobs

Job Composer

Your Jobs ▾ All Clusters ▾

Show  entries

Filter:

| ID                         | Name | User | Account | Time Used | Queue | Status | Cluster | Actions |
|----------------------------|------|------|---------|-----------|-------|--------|---------|---------|
| No data available in table |      |      |         |           |       |        |         |         |

Showing 0 to 0 of 0 entries

PreviousNext

powered by

 **Open OnDemand**

OnDemand version: 3.0.1

# COMPOSE A “JOB”

Open OnDemand / Job Composer

Jobs

Templates

? Help

## Jobs

+ New Job ▾

☆ Create Template

✎ Edit Files

⚙ Job Options

🔗 Open Terminal

▶ Submit

■ Stop

🗑 Delete

Show 25 ▾ entries

Search:

| Created                     | ↓↑ Name                         | ↓↑ ID | ↓↑ Cluster    | ↓↑ Status     | ↓↑ |
|-----------------------------|---------------------------------|-------|---------------|---------------|----|
| January 26, 2025<br>12:15pm | (default) Simple Sequential Job |       | SLURM Cluster | Not Submitted |    |

Showing 1 to 1 of 1 entries

Previous 1 Next

### Job Details

Job Name:

**(default) Simple Sequential Job**

Submit to:

SLURM Cluster

Account:

Not specified

Script location:

/home/databiom/ondemand/data/sys/myjobs/projects/default/1

# SHELL ACCESS



Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ 

>\_SLURM Cluster Shell Access

SLURM Cluster Shell Access

Your Jobs ▾ All Clusters ▾

## Active Jobs

Show  entries Filter:

| ID                         | Name | User | Account | Time Used | Queue | Status | Cluster | Actions |
|----------------------------|------|------|---------|-----------|-------|--------|---------|---------|
| No data available in table |      |      |         |           |       |        |         |         |

Showing 0 to 0 of 0 entries Previous Next

powered by



OnDemand version: 3.0.1

# TERMINAL OPENS IN NEW TAB

Host: localhost

databiom@localhost's password:

Last login: Sun Jan 26 11:57:54 2025 from [REDACTED]

[databiom@tsu-hpc01 ~]\$

# CHANGING YOUR PASSWORD

Host: localhost

databiom@localhost's password:

Last login: Sun Jan 26 11:57:54 2025 from 192.168.222.97

[databiom@tsu-hpc01 ~]\$ passwd

Changing password for user databiom.

Current Password:

New password:

Retype new password:

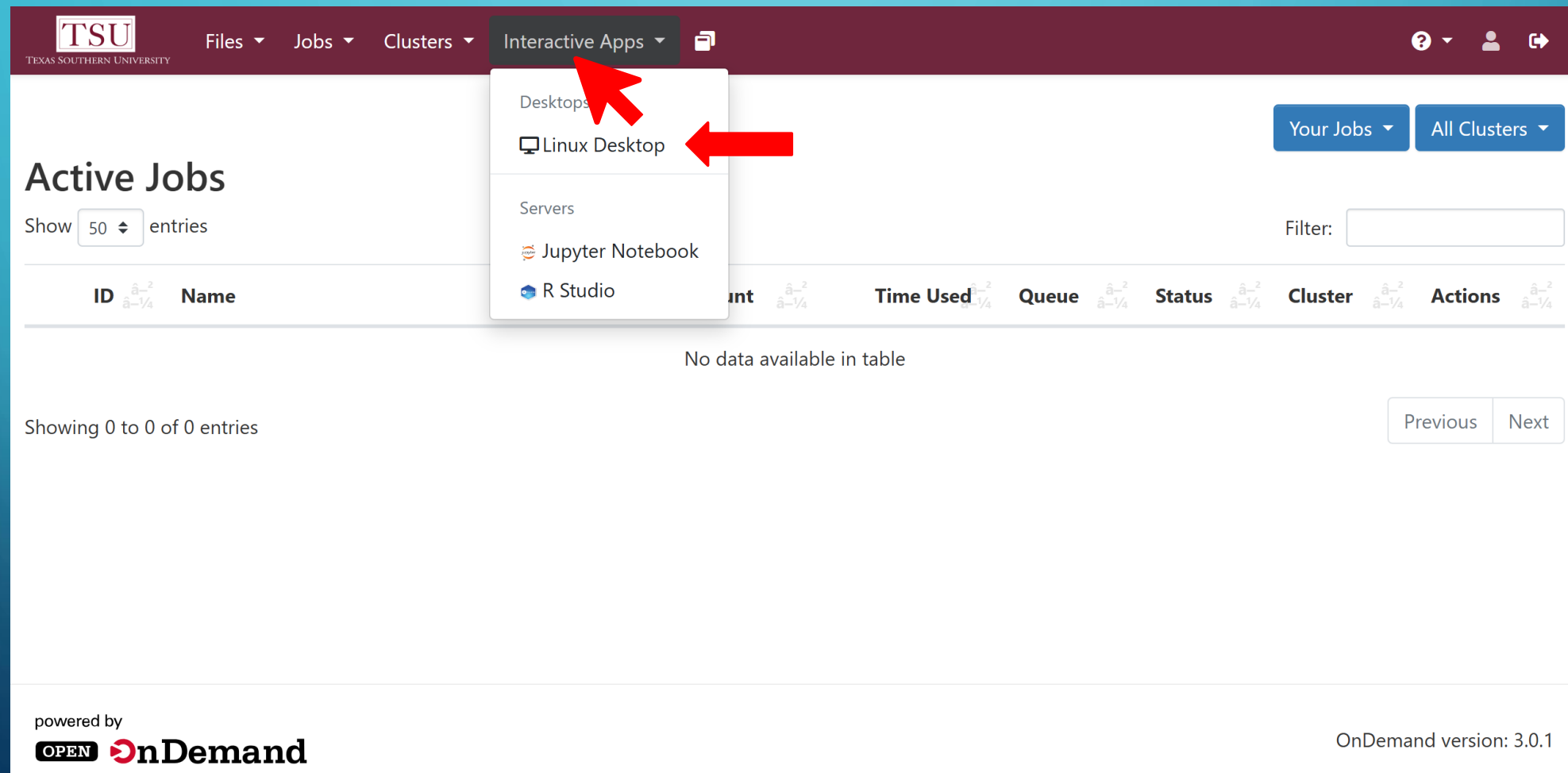
passwd: all authentication tokens updated successfully.

[databiom@tsu-hpc01 ~]\$



Text will not show up

# REQUESTING A LINUX DESKTOP SESSION



The screenshot shows the TSU OnDemand web interface. The top navigation bar is dark red with the TSU logo and links for Files, Jobs, Clusters, and Interactive Apps. The 'Interactive Apps' dropdown menu is open, showing options: Desktops, Linux Desktop (selected with a red arrow), Servers, Jupyter Notebook, and R Studio. The main content area is titled 'Active Jobs' and shows a table with columns: ID, Name, Amount, Time Used, Queue, Status, Cluster, and Actions. The table is currently empty, displaying 'No data available in table'. The footer includes the 'powered by OPEN OnDemand' logo and the version 'OnDemand version: 3.0.1'.

TSU  
TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾

Desktops  
Linux Desktop  
Servers  
Jupyter Notebook  
R Studio

Your Jobs ▾ All Clusters ▾

Active Jobs

Show 50 entries

Filter:

| ID                         | Name | Amount | Time Used | Queue | Status | Cluster | Actions |
|----------------------------|------|--------|-----------|-------|--------|---------|---------|
| No data available in table |      |        |           |       |        |         |         |

Showing 0 to 0 of 0 entries

Previous Next

powered by  
OPEN OnDemand

OnDemand version: 3.0.1

# REQUESTING A LINUX DESKTOP SESSION

The screenshot shows the TSU (Texas Southern University) web interface for requesting a Linux Desktop session. The top navigation bar includes links for Files, Jobs, Clusters, and Interactive Apps. The breadcrumb trail indicates the path: Home / My Interactive Sessions / Linux Desktop.

**Interactive Apps**

- Desktops
  - Linux Desktop**
- Servers
  - Jupyter Notebook
  - R Studio

**Linux Desktop**

This app will launch an interactive desktop on a compute node. You will have full access to the resources defined below. This is analogous to an interactive batch job.

How many resources do you need for this desktop?

- ☒ 4 Cores
- ☐ 8 Cores
- ☐ 16 Cores, 1 GPU
- ☐ 32 Cores, 1 GPU

How many hours do you need the resources for?

2

**Launch**

\* The Linux Desktop session data for this session can be accessed under the [data root directory](#).

# REQUESTING A LINUX DESKTOP SESSION

TSU  
TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ 

? ▾  

Session was successfully created. 

Home / My Interactive Sessions

Interactive Apps

Desktops

 Linux Desktop

Servers

 Jupyter Notebook

 R Studio

Linux Desktop (4201)

Queued

Created at: 2025-01-26 12:33:43 CST

Time Requested: 2 hours

Session ID: [cf54a055-29d2-411d-9a83-e3fa32633f77](#)

Please be patient as your job currently sits in queue. The wait time depends on the number of cores as well as time requested.

 Delete



powered by  
 

OnDemand version: 3.0.1

# LAUNCH LINUX DESKTOP SESSION



TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ 

 ▾  

Session was successfully created. 

Home / My Interactive Sessions

Interactive Apps

Desktops

 Linux Desktop

Servers

 Jupyter Notebook

 R Studio

Linux Desktop (4201) 1 node | 4 cores | Running 

Host: >\_gpu001.cm.cluster  Delete

Created at: 2025-01-26 12:33:43 CST

Time Remaining: 1 hour and 59 minutes

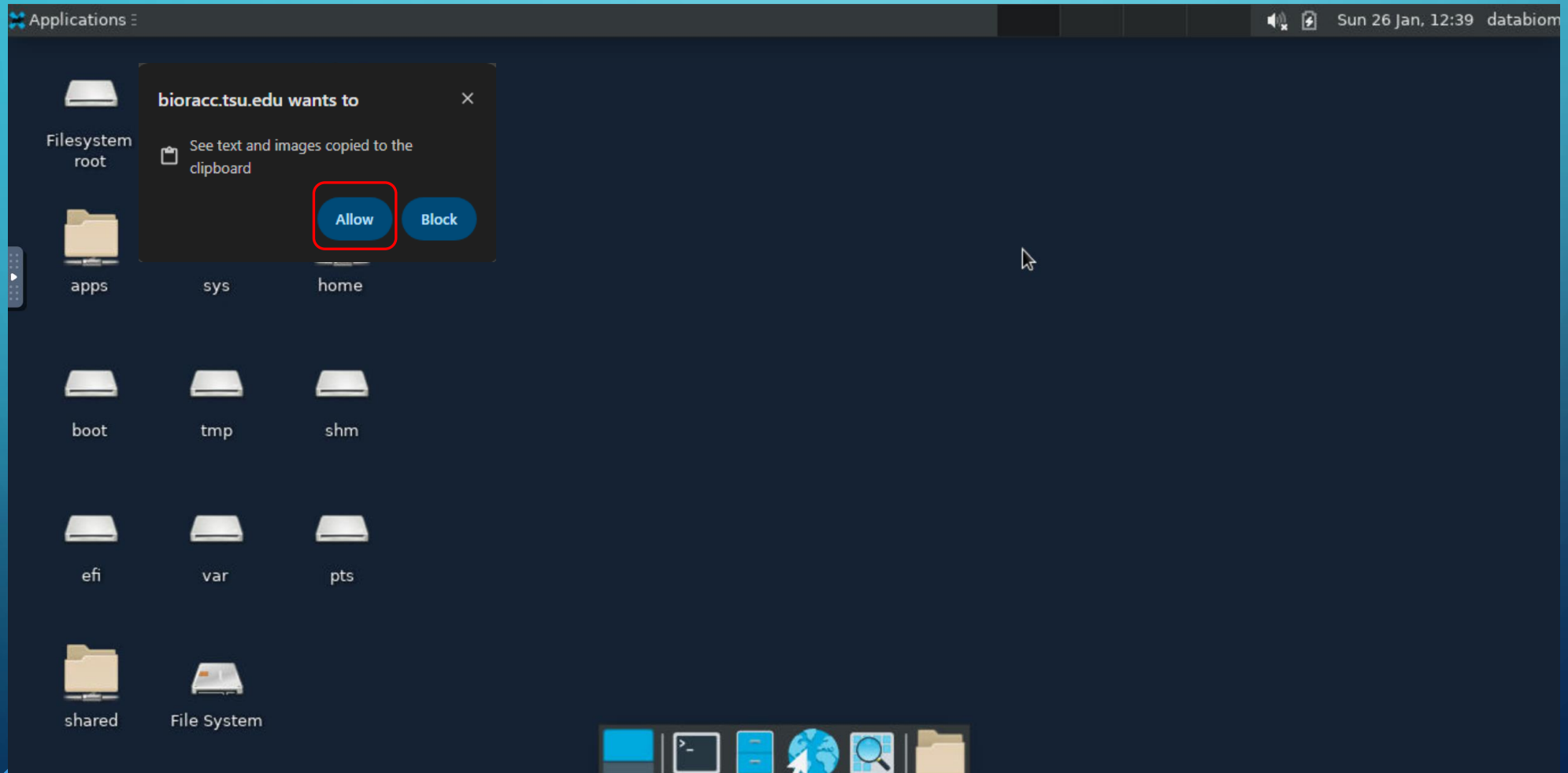
Session ID: cf54a055-29d2-411d-9a83-e3fa32633f77

Compression  0 (low) to 9 (high)

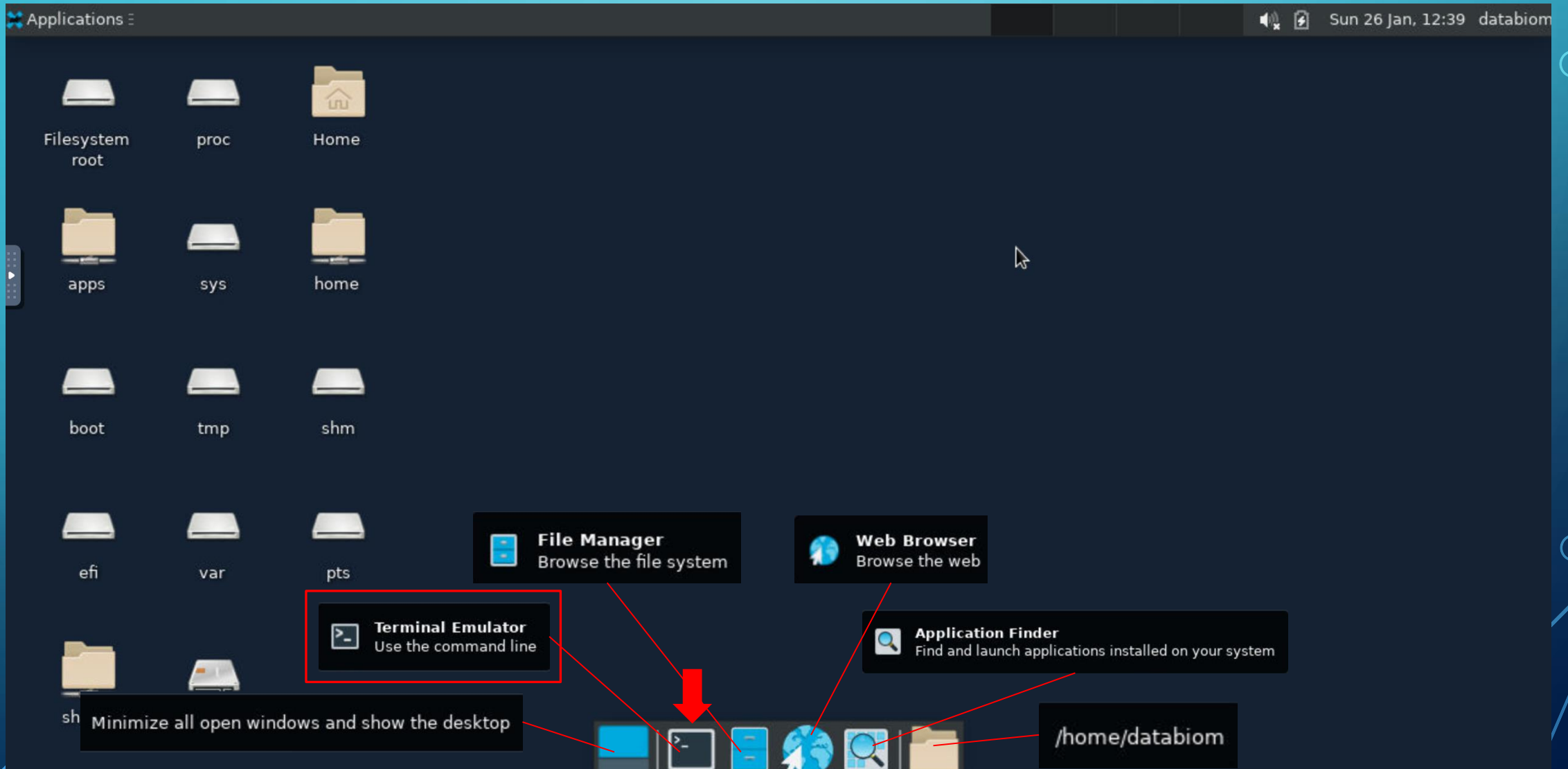
Image Quality  0 (low) to 9 (high)

 Launch Linux Desktop View Only (Share-able Link)

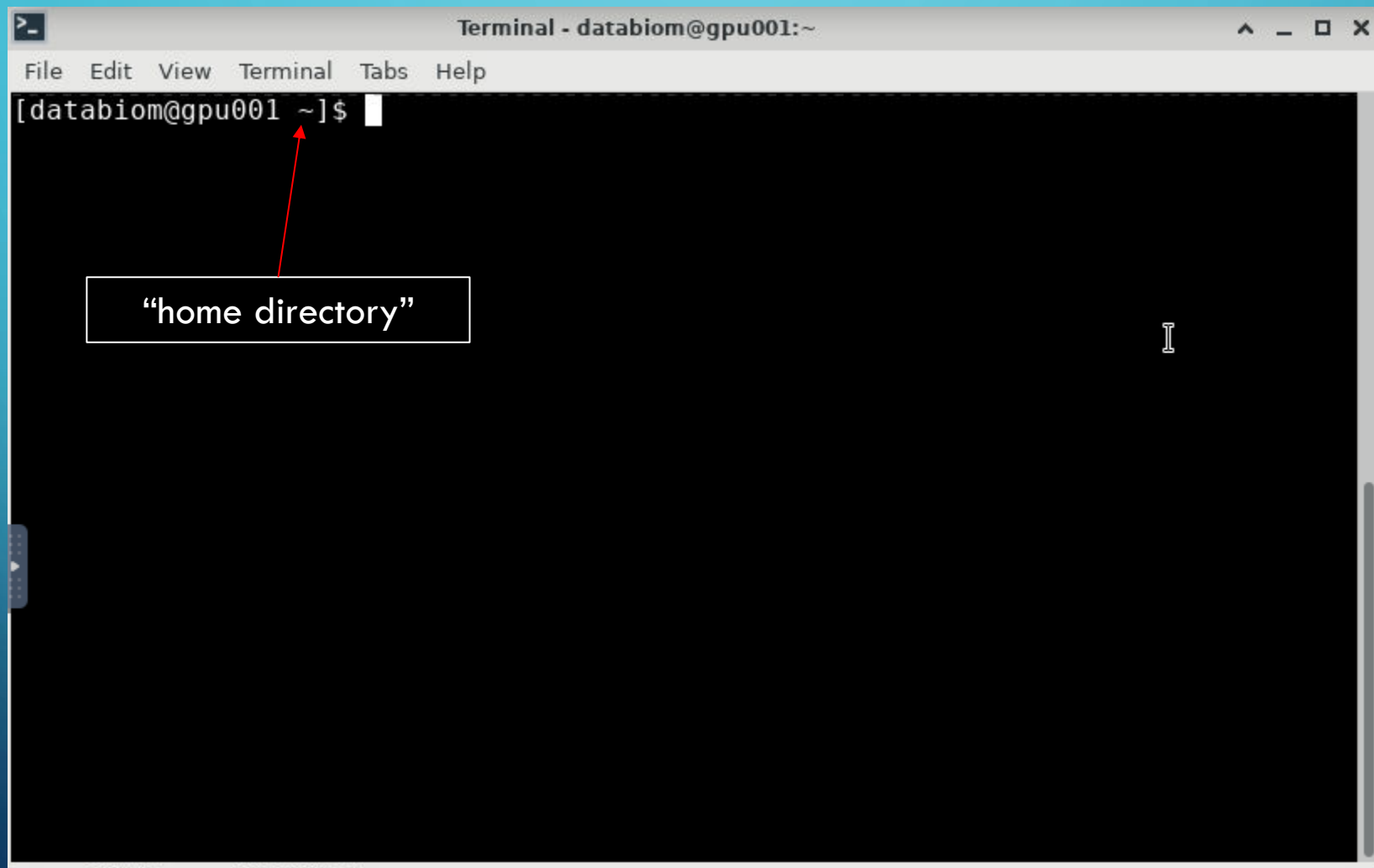
# ALLOW COPY/PASTE



# LINUX DESKTOP LAYOUT



# TERMINAL



A terminal window titled "Terminal - databiom@gpu001:~" is shown. The window has a menu bar with "File", "Edit", "View", "Terminal", "Tabs", and "Help". The terminal content shows the prompt "[databiom@gpu001 ~]\$" followed by a cursor. A red arrow points from a text box labeled "home directory" to the tilde symbol (~) in the prompt, indicating that ~ represents the home directory.

```
Terminal - databiom@gpu001:~  
File Edit View Terminal Tabs Help  
[databiom@gpu001 ~]$
```

home directory

# COMMON LINUX COMMANDS

- **ls** – “list” current working directory
- **cd** – “change directory”
  - **cd Directory1** → change/go to Directory1
  - **cd ..** → go back/up one directory
  - **cd** → will change/go to your home directory
- **mkdir** – “make directory”, create a new directory located in current directory
  - → **mkdir NewDirectory**

Everything is case sensitive!



# COMMON LINUX COMMANDS

- `cp` → “copy” a file
- `mv` → “move” a file



These use the syntax  
“*command file destination*”  
e.g.: `cp file1 Directory1`

- `rm` → “remove” a file



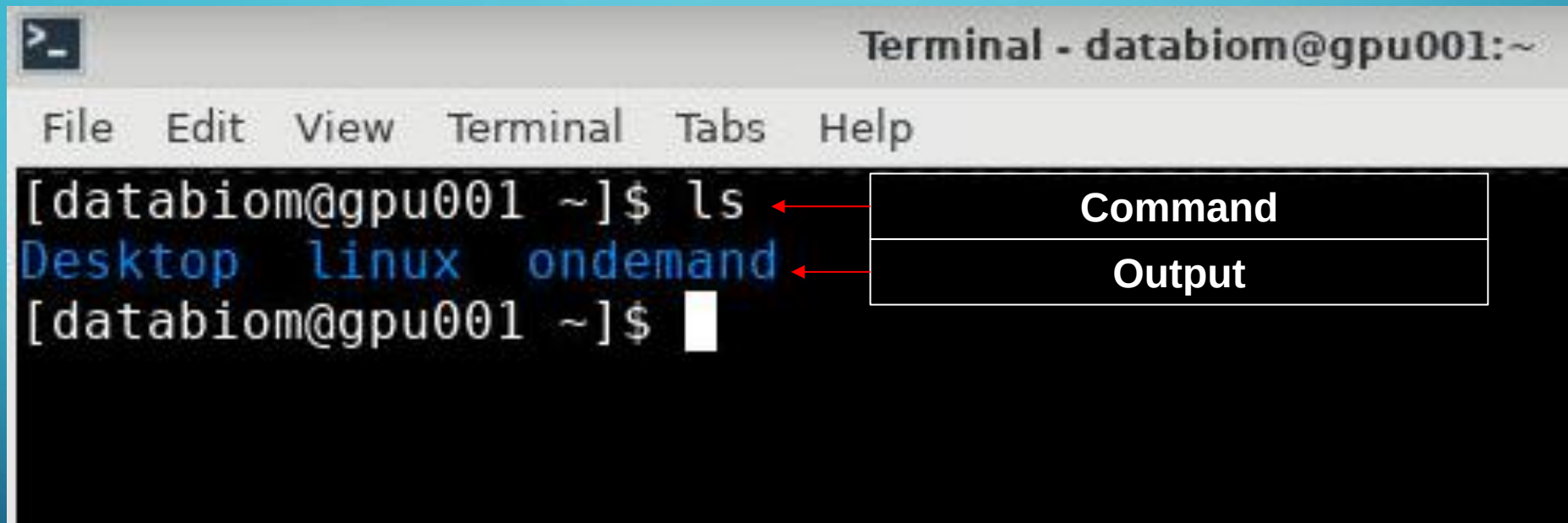
**\*\*There is no undo in linux\*\***

- `head` → view first few lines of a file
- `tail` → view last few lines of a file
- `less` → view a file one page at a time (good for large files)



# LIST CONTENTS OF CURRENT DIRECTORY

Everything is case sensitive!



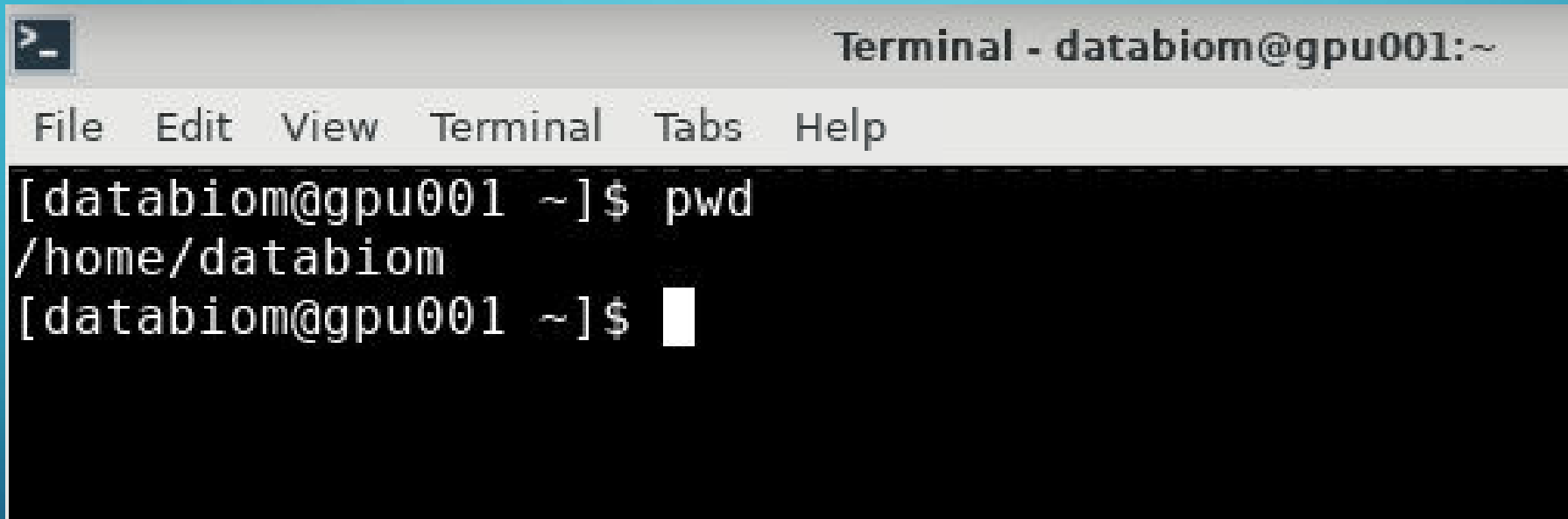
A terminal window titled "Terminal - databiom@gpu001:~" with a menu bar (File, Edit, View, Terminal, Tabs, Help). The prompt is "[databiom@gpu001 ~]\$". The command "ls" has been entered, and the output "Desktop linux ondemand" is displayed on the next line. A red arrow points from the "Command" label to the "ls" command, and another red arrow points from the "Output" label to the output text.

|         |
|---------|
| Command |
| Output  |

\*Hit 'Enter' to submit commands

# GET PATH OF WORKING DIRECTORY

Everything is case sensitive!

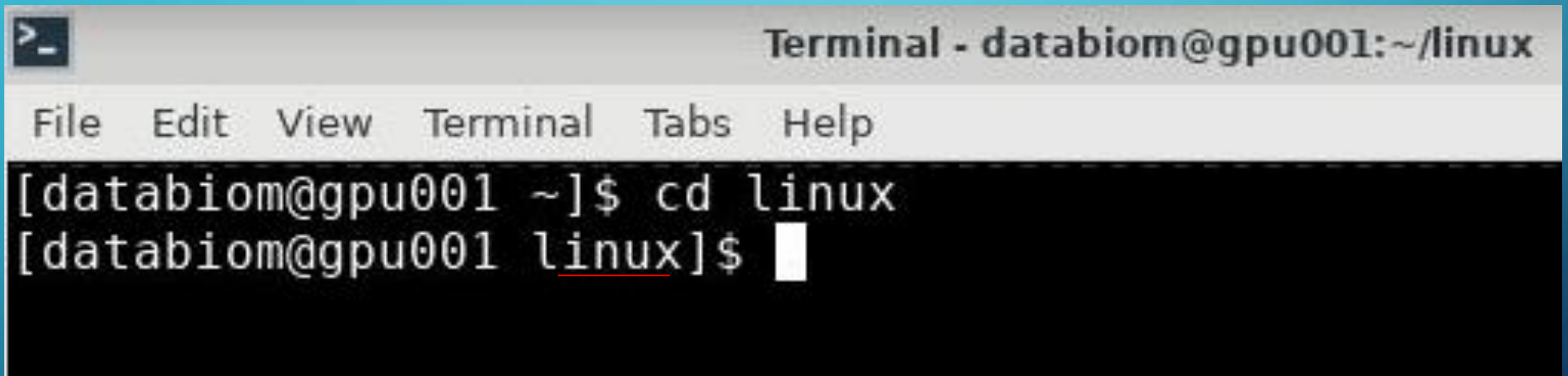
A terminal window titled "Terminal - databiom@gpu001:~" with a menu bar containing "File", "Edit", "View", "Terminal", "Tabs", and "Help". The terminal shows the command "[databiom@gpu001 ~]\$ pwd" and its output "/home/databiom". The prompt "[databiom@gpu001 ~]\$" is followed by a white cursor block.

```
Terminal - databiom@gpu001:~  
File Edit View Terminal Tabs Help  
[databiom@gpu001 ~]$ pwd  
/home/databiom  
[databiom@gpu001 ~]$
```

# MAKE A NEW DIRECTORY

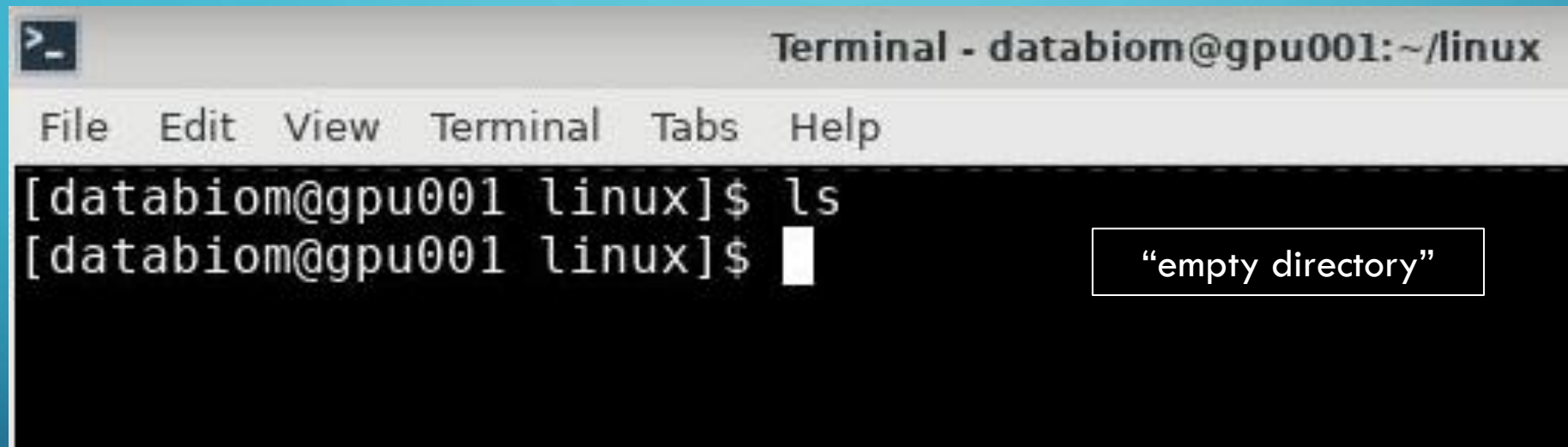
```
Terminal - databiom@gpu001:~  
File Edit View Terminal Tabs Help  
[databiom@gpu001 ~]$ mkdir linux  
[databiom@gpu001 ~]$ ls  
Desktop linux ondemand  
[databiom@gpu001 ~]$
```

# CHANGE (GO INTO A) DIRECTORY

A terminal window titled "Terminal - databiom@gpu001:~/linux" with a menu bar containing "File", "Edit", "View", "Terminal", "Tabs", and "Help". The terminal shows the command "cd linux" being executed, changing the directory from "~" to "linux". The prompt changes from "[databiom@gpu001 ~]" to "[databiom@gpu001 linux]", with "linux" underlined in red. A white cursor is visible after the second prompt.

```
>_ Terminal - databiom@gpu001:~/linux
File Edit View Terminal Tabs Help
[databiom@gpu001 ~]$ cd linux
[databiom@gpu001 linux]$
```

# LIST CONTENTS (OR LACK OF) IN DIRECTORY

A terminal window titled "Terminal - databiom@gpu001:~/linux" with a menu bar containing "File", "Edit", "View", "Terminal", "Tabs", and "Help". The terminal shows the command "ls" being executed, resulting in no output, which indicates an empty directory. A callout box points to the empty line after the command.

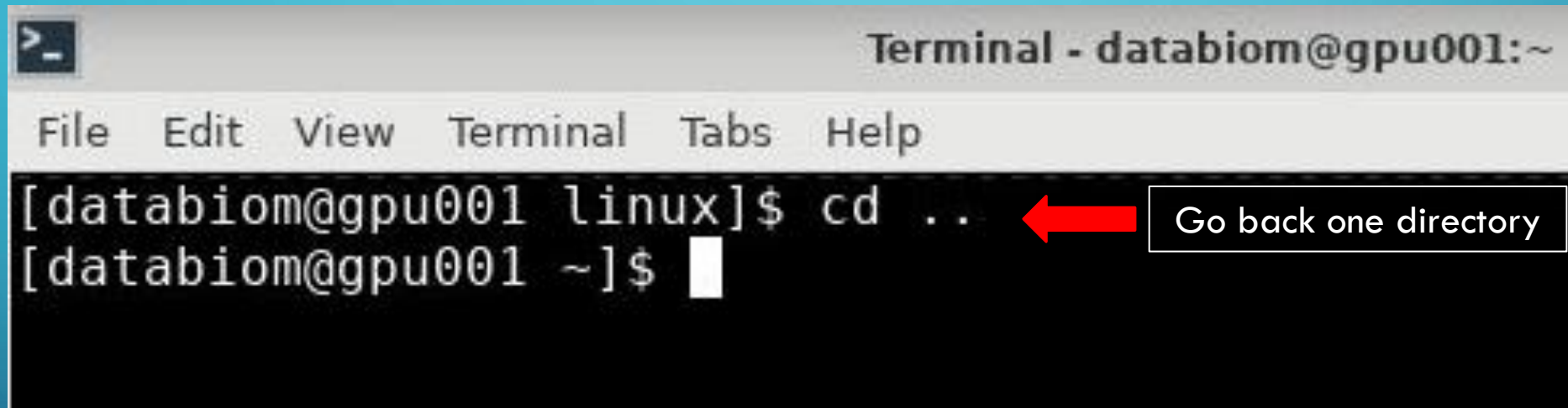
```
>_ Terminal - databiom@gpu001:~/linux
File Edit View Terminal Tabs Help
[databiom@gpu001 linux]$ ls
[databiom@gpu001 linux]$
```

"empty directory"

# DIRECTORY STRUCTURE/HIERARCHY

```
Terminal - databiom@gpu001:~/linux
File Edit View Terminal Tabs Help
[databiom@gpu001 linux]$ pwd
/home/databiom/linux
[databiom@gpu001 linux]$
```

# CHANGE DIRECTORY (GO BACK)

A terminal window titled "Terminal - databiom@gpu001:~" with a menu bar containing "File", "Edit", "View", "Terminal", "Tabs", and "Help". The terminal shows a command prompt where the user enters "cd ..". A red arrow points from a text box to the ".." in the command. The prompt then changes to show the user is in the home directory (~).

```
>_ Terminal - databiom@gpu001:~  
File Edit View Terminal Tabs Help  
[databiom@gpu001 linux]$ cd ..  
[databiom@gpu001 ~]$
```

Go back one directory

"cd" followed by nothing will take you back to your home directory

# USING THE WGET COMMAND TO DOWNLOAD FROM INTERNET

```
Terminal - databiom@gpu011:~/linux
File Edit View Terminal Tabs Help
[databiom@gpu011 ~]$ cd linux/
[databiom@gpu011 linux]$ wget https://raw.githubusercontent.com/ScottWidmann/databiom/refs/heads/main/breast_cancer.txt
```

**<https://raw.githubusercontent.com/ScottWidmann/databiom/refs/heads/main/>**

# DOWNLOAD INTO YOUR LINUX DIRECTORY

```
Terminal - databiom@gpu011:~/linux
File Edit View Terminal Tabs Help
[databiom@gpu011 ~]$ cd linux/
[databiom@gpu011 linux]$ wget https://raw.githubusercontent.com/ScottWidmann/databiom/refs/heads/main/breast_cancer.txt
--2025-01-26 17:10:52-- https://raw.githubusercontent.com/ScottWidmann/databiom/refs/heads/main/breast_cancer.txt
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.133, 185.199.110.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 11366591 (11M) [text/plain]
Saving to: 'breast_cancer.txt.2'

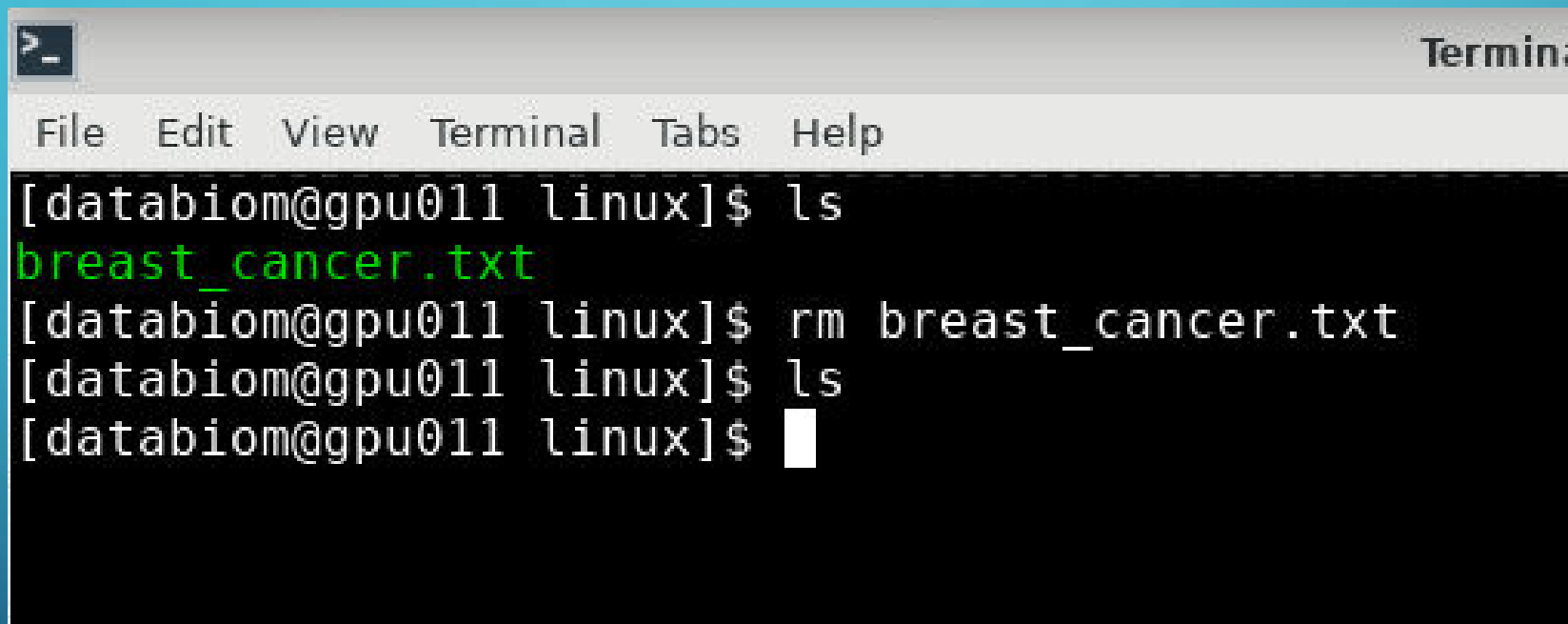
breast_cancer.txt.2      100%[=====>] 10.84M  67.7MB/s   in 0.2s

2025-01-26 17:10:52 (67.7 MB/s) - 'breast_cancer.txt.2' saved [11366591/11366591]

[databiom@gpu011 linux]$
```

“clear” will remove everything above if you want

# REMOVING FILES

A terminal window titled 'Terminal' with a menu bar containing 'File', 'Edit', 'View', 'Terminal', 'Tabs', and 'Help'. The terminal shows a user named 'databiom' on a host named 'gpu011' in a 'linux' directory. The user runs 'ls' and sees 'breast\_cancer.txt' in green. Then they run 'rm breast\_cancer.txt'. Finally, they run 'ls' again and the prompt returns without any output, indicating the file has been removed.

```
[databiom@gpu011 linux]$ ls
breast_cancer.txt
[databiom@gpu011 linux]$ rm breast_cancer.txt
[databiom@gpu011 linux]$ ls
[databiom@gpu011 linux]$
```

# COPYING FILES

```
Terminal - databiom@gpu001:~  
File Edit View Terminal Tabs Help  
[databiom@gpu001 linux]$ cd ..  
[databiom@gpu001 ~]$ cp /home/databiom/linux/breast_cancer.txt .
```

Syntax: “*command* *[file]* *[destination]*”  
e.g.: cp file1 Directory1



“copy to current  
directory”

# MOVING FILES

```
Terminal -
File Edit View Terminal Tabs Help
[databiom@gpu011 ~]$ ls
breast_cancer.txt Desktop linux ondemand
[databiom@gpu011 ~]$ mv breast_cancer.txt linux/
[databiom@gpu011 ~]$ cd linux/
[databiom@gpu011 linux]$ ls
breast_cancer.txt
[databiom@gpu011 linux]$
```

**Syntax: “*command [file] [destination]*”**  
e.g.: **mv file1 Directory1**

# USING A TEXT EDITOR TO READ AND EDIT FILES

```
>_
File Edit View Terminal Tabs Help
[databiom@gpu011 linux]$ nano breast_cancer.txt
```

Text editor

Text file

```
Terminal - databiom@gpu011:~/linux
GNU nano 2.9.8 breast_cancer.txt
Hugo_Symbol  Entrez_Gene_Id  Center  NCBI_Build  Chromosome  Start_Position  End_Position  Strand  Variant_Classif$
AKT1  207  MSKCC  GRCh37  14  105246551  105246551  +  Missense_Mutation  SNP  C  C  $
FOXA1  3169  MSKCC  GRCh37  14  38061240  38061240  +  Missense_Mutation  SNP  G  G  $
FOXA1  3169  MSKCC  GRCh37  14  38061240  38061240  +  Missense_Mutation  SNP  G  G  $
DICER1  23405  MSKCC  GRCh37  14  95574698  95574698  +  Missense_Mutation  SNP  C  C  $
AKT1  207  MSKCC  GRCh37  14  105246551  105246551  +  Missense_Mutation  SNP  C  C  $
DICER1  23405  MSKCC  GRCh37  14  95562249  95562249  +  Missense_Mutation  SNP  C  C  $
DICER1  23405  MSKCC  GRCh37  14  95562693  95562693  +  Missense_Mutation  SNP  C  C  $
DICER1  23405  MSKCC  GRCh37  14  95562249  95562249  +  Missense_Mutation  SNP  C  C  $
DICER1  23405  MSKCC  GRCh37  14  95562693  95562693  +  Missense_Mutation  SNP  C  C  $
FOXA1  3169  MSKCC  GRCh37  14  38060789  38060800  +  In_Frame_Del  DEL  GAACGGGTGGTT  GAACGGGT$
FOXA1  3169  MSKCC  GRCh37  14  38061188  38061189  +  In_Frame_Ins  INS  -  CTT  $
AX  4149  MSKCC  GRCh37  14  65544679  65544679  +  Nonsense_Mutation  SNP  G  G  $
AKT1  207  MSKCC  GRCh37  14  105246551  105246551  +  Missense_Mutation  SNP  C  C  $
NKX2-1  7080  MSKCC  GRCh37  14  36987093  36987093  +  Missense_Mutation  SNP  G  G  $
FOXA1  3169  MSKCC  GRCh37  14  38060953  38060953  +  Missense_Mutation  SNP  C  C  $
FOXA1  3169  MSKCC  GRCh37  14  38060986  38060986  +  Missense_Mutation  SNP  G  G  $
AKT1  207  MSKCC  GRCh37  14  105246551  105246551  +  Missense_Mutation  SNP  C  C  $
AKT1  207  MSKCC  GRCh37  14  105246551  105246551  +  Missense_Mutation  SNP  C  C  $
AKT1  207  MSKCC  GRCh37  14  105246551  105246551  +  Missense_Mutation  SNP  C  C  $
AKT1  207  MSKCC  GRCh37  14  105258970  105258970  +  Missense_Mutation  SNP  A  A  $
FOXA1  3169  MSKCC  GRCh37  14  38061231  38061231  +  Missense_Mutation  SNP  A  A  $
MAX  4149  MSKCC  GRCh37  14  65544703  65544703  +  Nonsense_Mutation  SNP  G  G  $
MAX  4149  MSKCC  GRCh37  14  65544703  65544703  +  Nonsense_Mutation  SNP  G  G  $
AKT1  207  MSKCC  GRCh37  14  105246436  105246436  +  Missense_Mutation  SNP  A  A  $
FOXA1  3169  MSKCC  GRCh37  14  38061007  38061022  +  Frame_Shift_Del  DEL  GCCCGGGGGCCGGCGC  $
AKT1  207  MSKCC  GRCh37  14  105246551  105246551  +  Missense_Mutation  SNP  C  C  $
AKT1  207  MSKCC  GRCh37  14  105246551  105246551  +  Missense_Mutation  SNP  C  C  $
AKT1  207  MSKCC  GRCh37  14  105239673  105239673  +  Missense_Mutation  SNP  G  G  $
RAD51B  5890  MSKCC  GRCh37  14  68331719  68331719  +  Splice_Site  SNP  G  G  T  $
AKT1  207  MSKCC  GRCh37  14  105246551  105246551  +  Missense_Mutation  SNP  C  C  $
AKT1  207  MSKCC  GRCh37  14  105246551  105246551  +  Missense_Mutation  SNP  C  C  $
[ Read 10196 lines (Converted from DOS format) ]
^G Get Help  ^O Write Out  ^W Where Is  ^K Cut Text  ^J Justify  ^C Cur Pos  M-U Undo  M-A Mark Text
^X Exit      ^R Read File  ^\ Replace   ^U Uncut Text ^T To Spell  ^_ Go To Line M-E Redo    M-6 Copy Text
```

ctrl + x to exit

# GREP COMMAND FOR FILTERING

## “Pattern of interest”

## “Filename”



A screenshot of a terminal window. The title bar at the top says "Terminal". Below the title bar is a menu bar with the options "File", "Edit", "View", "Terminal", "Tabs", and "Help". The terminal prompt is "[databiom@gpu011 linux]\$". The command being entered is "grep 'Y537S' breast\_cancer.txt". Two red arrows point to the command: one points to the single quote character before 'Y537S', and the other points to the file name 'breast\_cancer.txt'.

```
[databiom@gpu011 linux]$ grep "Y537S" breast_cancer.txt
```

## “Pattern found”

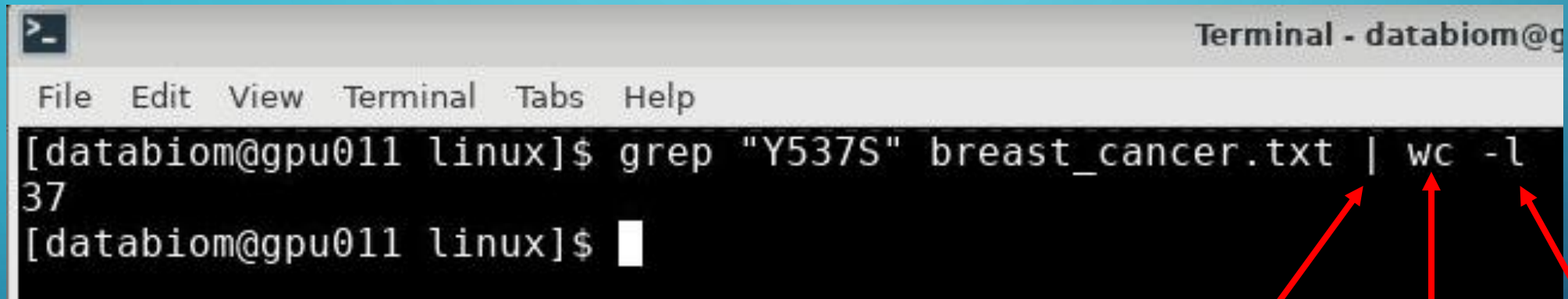
```

Terminal - databiom@gpu011:~/linux
File Edit View Terminal Tabs Help

A          "1,1"          NA      NA      NA      NA
ESR1      2099      MSKCC      GRCh37      6      152419923      152419923      +      Missense_Mutation      SNP      A      A      C
NA      P-0014995-T01-IM6      NORMAL      A      A      NA      NA      NA      NA      NA      Unknown SOMATIC      NA      NA      N
A      MSK-IMPACT      NA      NA      NA      NA      c.1610A>C      p.Tyr537Ser      p.Y537S      ENST00000440973      10/10      555      4
51      104      391      391      0      "ESR1,missense_variant,p.Tyr537Ser,ENST00000440973,NM_001122742.1;ESR1,missense_variant,p
.Tyr537Ser,ENST00000206249,NM_000125.3;ESR1,missense_variant,p.Tyr537Ser,ENST00000443427,NM_001122740.1;ESR1,missense_variant,p.T
yr425Ser,ENST00000456483,NM_001122741.1;ESR1,missense_variant,p.Tyr537Ser,ENST00000338799,;ESR1,missense_variant,p.Tyr276Ser,ENST
00000406599,;ESR1,intron_variant,ENST00000427531,;"      C      ENSG00000091831      ENST00000440973      Transcript      missense_variant1
980/6466      1610/1788      537/595      Y/S      tAt/tCt      "COSM1074637,COSM1074639"      1      NA      ESR1      HGNC      3467      p
rotein_coding      YES      CCDS5234.1      ENSP00000405330      P03372      "Q9UE35,Q9UBT1,Q5T5H8,K7R989,K4Q331,K4PXX0,K4PX46,G4XH65,E3WH18,C
8CJL0,C7FFR5,B6DU67"      UPI000004A328      NM_001122742.1      deleterious(0)      probably_damaging(0.996)      "Gene3D:1.10.565.
10,PIRSF_domain:PIRSF002527,PIRSF_domain:PIRSF500101,hmmpanther:PTHR24084,hmmpanther:PTHR24084:SF3,Superfamily_domains:SSF48508"N
A          "1,1"          NA      NA      NA      NA
ESR1      2099      MSKCC      GRCh37      6      152419923      152419923      +      Missense_Mutation      SNP      A      A      C
NA      P-0015097-T01-IM6      NORMAL      A      A      NA      NA      NA      NA      NA      Unknown SOMATIC      NA      NA      N
NA      MSK-IMPACT      NA      NA      NA      NA      c.1610A>C      p.Tyr537Ser      p.Y537S      ENST00000440973      10/10      646      5
96      50      391      391      0      "ESR1,missense_variant,p.Tyr537Ser,ENST00000440973,NM_001122742.1;ESR1,missense_variant,p
.Tyr537Ser,ENST00000206249,NM_000125.3;ESR1,missense_variant,p.Tyr537Ser,ENST00000443427,NM_001122740.1;ESR1,missense_variant,p.T

```

# “PIPING” TO OTHER COMMANDS



A terminal window titled "Terminal - databiom@g" with a menu bar (File, Edit, View, Terminal, Tabs, Help). The command `grep "Y537S" breast_cancer.txt | wc -l` is entered. The output `37` is displayed on the line below. The prompt `[databiom@gpu011 linux]$` is shown again on the next line.

```
Terminal - databiom@g
File Edit View Terminal Tabs Help
[databiom@gpu011 linux]$ grep "Y537S" breast_cancer.txt | wc -l
37
[databiom@gpu011 linux]$
```

“Pipe”

“Word count”

“Count lines”

## REDIRECT OUTPUT TO A NEW FILE



A terminal window titled "Terminal - databiom@gpu011:" with a menu bar (File, Edit, View, Terminal, Tabs, Help). The command prompt shows the user running `grep "Y537S" breast_cancer.txt > Y537S.txt` and then `nano Y537S.txt`. Two red arrows point to the output file `Y537S.txt` in the command.

```
Terminal - databiom@gpu011:
File Edit View Terminal Tabs Help
[databiom@gpu011 linux]$ grep "Y537S" breast_cancer.txt > Y537S.txt
[databiom@gpu011 linux]$ nano Y537S.txt
```

[illegible]

“new file”

“redirect”

[illegible]

```
Terminal - databiom
File Edit View Terminal Tabs Help
[databiom@gpu011 linux]$ sed -i 's/ESR1/ERalpha/' Y537S.txt
```

**“in place”      “substitute”      “old”      “new”**

[illegible]

# EXTRACTING COLUMNS WITH AWK COMMAND

```
Terminal - databiom@gpu001:~/linux
File Edit View Terminal Tabs Help
[databiom@gpu001 linux]$ awk '{print $9}' breast_cancer.txt > mutation_types.txt
[databiom@gpu001 linux]$ nano mutation_types.txt
```

“column number”

```
Terminal - databiom@gpu001:~/linux
File Edit View Terminal Tabs Help
GNU nano 2.9.8 mutation_types.txt
Variant_Classification
Missense_Mutation
Missense_Mutation
Missense_Mutation
Missense_Mutation
Missense_Mutation
Missense_Mutation
Missense_Mutation
Missense_Mutation
Missense_Mutation
Missense_Mutation
In_Frame_Del
In_Frame_Ins
onsense_Mutation
issense_Mutation
Missense_Mutation
Missense_Mutation
```

# MODULES (PROGRAMS/APPLICATIONS)

Type 'module avail' to see all the available modules

```
(base) ~]$ module avail

----- /cm/local/modulefiles -----
boost/1.77.0      cmd      freeipmi/1.6.8  luajit      null      shared
cluster-tools-dell/9.2  cmjob    gcc/11.2.0     mariadb-libs  openldap  slurm/slurm/21.08.8
cluster-tools/9.2      cuda-dcgm/2.4.6.1  ipmitool/1.8.18  module-git   python3
cm-bios-tools          dot      lua/5.4.4      module-info  python39

----- /cm/shared/modulefiles -----
blacs/openmpi/gcc/64/1.1patch03  default-environment  intel/compiler/64/2020/19.1.3
blas/gcc/64/3.10.0               fftw3/openmpi/gcc/64/3.3.10  intel/mpi/64/(default)
bonnie++/2.00a                   gdb/10.2                 intel/mpi/64/2020/4.304
cm-pmix3/3.1.4                   globalarrays/openmpi/gcc/64/5.8  iozone/3_492
cm-pmix4/4.1.1                   hdf5/1.12.1               jupyter-eg-kernel-wlm-py39/3.0.1
cuda11.0/blas/11.0.3              hdf5_18/1.8.21             jupyter/14.2.2
cuda11.0/fft/11.0.3                hwloc/1.11.11              lapack/gcc/64/3.10.0
cuda11.0/toolkit/11.0.3             hwloc2/2.7.1               mpich/ge/gcc/64/3.4.2
cuda11.8/blas/11.8.0               intel-cluster-runtime/ia32/2019.6  mvapich2/gcc/64/2.3.7
cuda11.8/fft/11.8.0                intel-cluster-runtime/ia32/2020.4(default)  namd/20220721
cuda11.8/toolkit/11.8.0             intel-cluster-runtime/intel64/(default)  netcdf/gcc/64/gcc/64/4.8.1
cuda12.0/blas/12.0.1                intel-cluster-runtime/intel64/2019.6  netperf/2.7.0
cuda12.0/fft/12.0.1                 intel-cluster-runtime/intel64/2020.4(default)  openblas/dynamic/(default)
cuda12.0/toolkit/12.0.1              intel-tbb-oss/ia32/2021.4.0  openblas/dynamic/0.3.18
cudnn8.6-cuda11.8/8.6.0.163          intel-tbb-oss/intel64/2021.4.0  openmpi/gcc/64/4.1.2
cudnn8.8-cuda12.0/8.8.1.3            intel/compiler/32/2020/19.1.3  openmpi4/gcc/4.1.2
cudnn8.9-cuda12.0/8.9.6.50           intel/compiler/64/(default)    ucx/1.10.1

----- /apps/modulefiles -----
acpype/2022.7.21      cudnn/8.8.1.3      nextflow/23.04.02  rdkit/2023.3.2
adfrsuite/1.0         fastqc/0.12.1      nglview/3.0.6      ribodetector/0.2.7
amber/22              gcc/12.3.0          openbabel/2.4.0    RSEM/1.3.3
anaconda/2023.03      git/2.41.0          openeye/2023.2.3   samtools/1.17
apptainer/1.1.9       grace/5.1.25        openmm/8.0.0        sas/9.4
autodock_gpu/1.5.1    gromacs/2023.1      parallel/20230722  scikit-learn/1.2.2
autodock_vina/1.1.2   gromacs_gpu/2023.1  perl/5.36.1         smog2/2.4.5
boost/1.82.0           gsea/4.3.2          ptmsi/py-3.9        sratoolkit/3.0.5
bowtie/2.5.1           gtk3/3.24.38        ptmsi/py-3.9-age    star/2.7.10b
bwa-mem2/2.2.1         htop/3.2.2          ptmsi/ape/py-3.9    tensorflow-gpu/2.11.0-python-3.9.16
chimeraX/1              lapack/3.11.0       python/3.11.2        torch/1.12.1-py3.9
chimeraX/1.6.1          libcusolver/11.4.4.55  QT/5.15.3           torch/2.0.1-py3.9
chimeraX/1.8            mgltools/1.5.7       qt/5.15.3            vmd/1.9.4
```

Need a particular program installed?  
Request an installation.

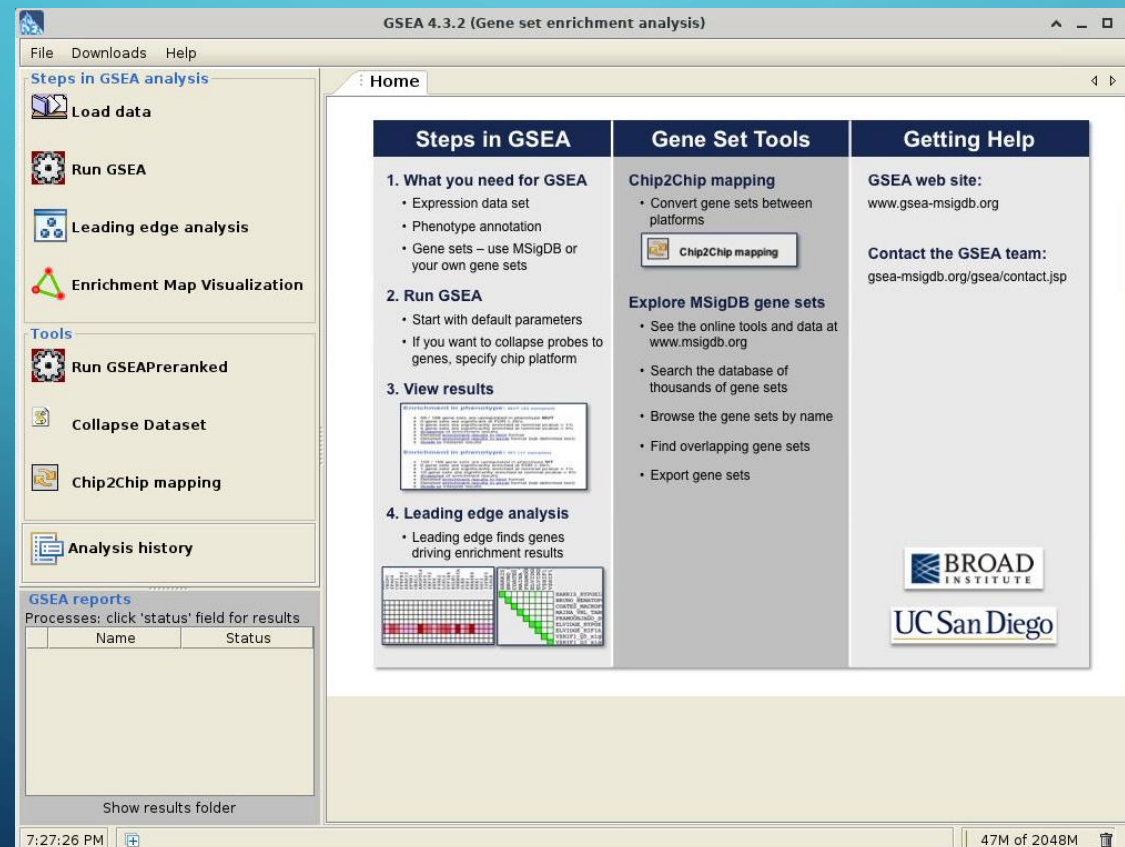
# RUNNING A MODULE

```
Terminal -
File Edit View Terminal Tabs Help
[databiom@gpu001 linux]$ module load gsea
[databiom@gpu001 linux]$ gsea.sh
```

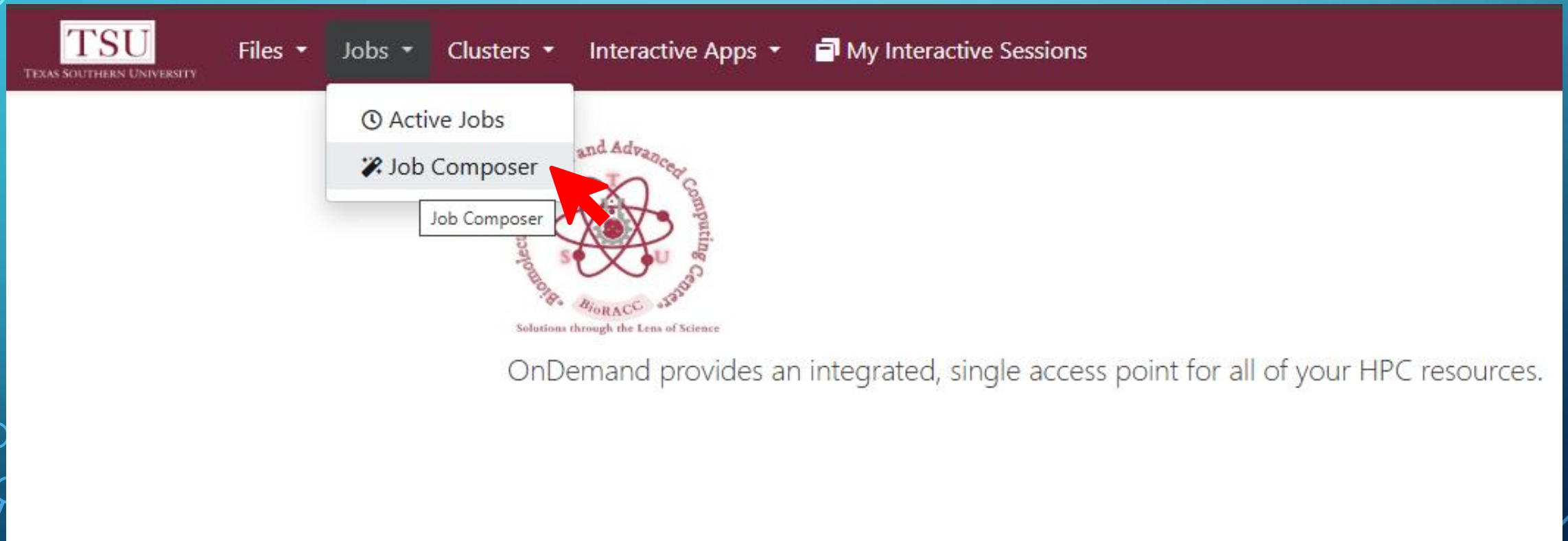
Must first load  
module

This particular app  
is started like this

Graphical User  
Interface (GUI)  
opens up



# SUBMITTING A JOB



The screenshot displays the OnDemand web interface for Texas Southern University (TSU). The top navigation bar is dark red and contains the TSU logo, a "Files" dropdown, a "Jobs" dropdown, a "Clusters" dropdown, an "Interactive Apps" dropdown, and a "My Interactive Sessions" link. The "Jobs" dropdown menu is open, showing two options: "Active Jobs" with a clock icon and "Job Composer" with a pencil icon. A red arrow points to the "Job Composer" option. Below the menu, there is a circular logo for the "BioRACC and Advanced Computing Center" with the text "Solutions through the Lens of Science". Below the logo, a text block states: "OnDemand provides an integrated, single access point for all of your HPC resources."

TSU  
TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ My Interactive Sessions

⌚ Active Jobs  
✎ Job Composer

Job Composer

BioRACC and Advanced Computing Center  
Solutions through the Lens of Science

OnDemand provides an integrated, single access point for all of your HPC resources.

# CREATE A NEW JOB

[Open OnDemand](#) / [Job Composer](#) [Jobs](#) [Templates](#)

Job was successfully destroyed.

## Jobs

+ New Job ▾

From Default Template

From Specified Path

From Selected Job

☆ Create Template

Open Terminal

Submit

Stop

Delete

Search:

| Created                     | Name | ID | Cluster | Status |
|-----------------------------|------|----|---------|--------|
| No data available in table  |      |    |         |        |
| Showing 0 to 0 of 0 entries |      |    |         |        |

Previous

Next

# NEW JOB

Open OnDemand / Job Composer Jobs Templates

Help

Job was successfully created.

## Jobs

+ New Job

☆ Create Template

Edit Files

Job Options

Open Terminal

Submit

Stop

Delete

Show 25 entries

Search:

| Created                 | Name                            | ID | Cluster       | Status        |
|-------------------------|---------------------------------|----|---------------|---------------|
| January 26, 2025 7:33pm | (default) Simple Sequential Job |    | SLURM Cluster | Not Submitted |

Showing 1 to 1 of 1 entries

Previous 1 Next

### Job Details

Job Name:

(default) Simple Sequential Job

Submit to:

SLURM Cluster

Account:

Not specified

Script location:

/home/databiom/ondemand/data/sys/myjobs/projects/default/1

Script name:

main\_job.sh

Folder Contents:

main\_job.sh

# JOB DETAILS

Job Details

Job Name:

**(default) Simple Sequential Job**

Submit to:

SLURM Cluster

Account:

Not specified

Script location:

/home/databiom/ondemand/data/sys/myjobs/projects/default/1

Script name:

main\_job.sh

Folder Contents:

main\_job.sh

Submit Script

main\_job.sh

Script contents:

```
#!/bin/bash
# JOB HEADERS HERE

echo "Hello World"
```

Script that will be  
executed

Prints "Hello World"

# SUBMIT JOB

Job was successfully created.

## Jobs

+ New Job ▾

☆ Create Template

📄 Edit Files

⚙️ Job Options

🔧 Open Terminal

Submit Job

▶ Submit

■ Stop

🗑 Delete

Show 25 ▾ entries

Search:

Created



Name



ID



Cluster



Status



January 26, 2025 7:45pm

(default) Simple Sequential Job

SLURM Cluster

Not Submitted

Showing 1 to 1 of 1 entries

Previous

1

Next

# JOB RUNNING

## Jobs

+ New Job ▾

☆ Create Template

Edit Files

⚙ Job Options

🔗 Open Terminal

▶ Submit

■ Stop

🗑 Delete

Show 25 ▾ entries

Search:

| Created                 | Name                            | ID   | Cluster       | Status  |
|-------------------------|---------------------------------|------|---------------|---------|
| January 26, 2025 7:45pm | (default) Simple Sequential Job | 4215 | SLURM Cluster | Running |

Showing 1 to 1 of 1 entries

Previous 1 Next

# JOB COMPLETED

## Jobs

+ New Job ▾

☆ Create Template

✎ Edit Files

⚙ Job Options

🔗 Open Terminal

▶ Submit

■ Stop

🗑 Delete

Show 25 ▾ entries

Search:

| Created                 | Name                            | ID   | Cluster       | Status    |
|-------------------------|---------------------------------|------|---------------|-----------|
| January 26, 2025 7:45pm | (default) Simple Sequential Job | 4215 | SLURM Cluster | Completed |

Showing 1 to 1 of 1 entries

Previous 1 Next

# CHECK OUTPUT OF JOB

## Jobs

+ New Job ▾

☆ Create Template

Open Directory in File Manager

Edit Files

Job Options

Open Terminal

Submit

Stop

Delete

Show 25 ▾ entries

Search:

| Created                 | Name                            | ID   | Cluster       | Status    |
|-------------------------|---------------------------------|------|---------------|-----------|
| January 26, 2025 7:45pm | (default) Simple Sequential Job | 4215 | SLURM Cluster | Completed |

Showing 1 to 1 of 1 entries

Previous

1

Next

# OUTPUT OF JOB

↑ / home / databiom / ondemand / data / sys / myjobs / projects / default / 1 / [Change directory](#)

| Type                                                                                                       | Name           | Size     |
|------------------------------------------------------------------------------------------------------------|----------------|----------|
| <input type="checkbox"/>  | main_job.sh    | 51 Bytes |
| <input type="checkbox"/>  | slurm-4215.out | 12 Bytes |

Hello World

# PUBLISHED DATASET ON GEO

  
Gene Expression Omnibus

HOMESEARCHSITE MAPGEO PublicationsFAQMIAMEEmail GEO

NCBI > GEO > **Accession Display** [?](#)Not logged in | [Login](#) [?](#)

Scope:  Format:  Amount:  GEO accession:

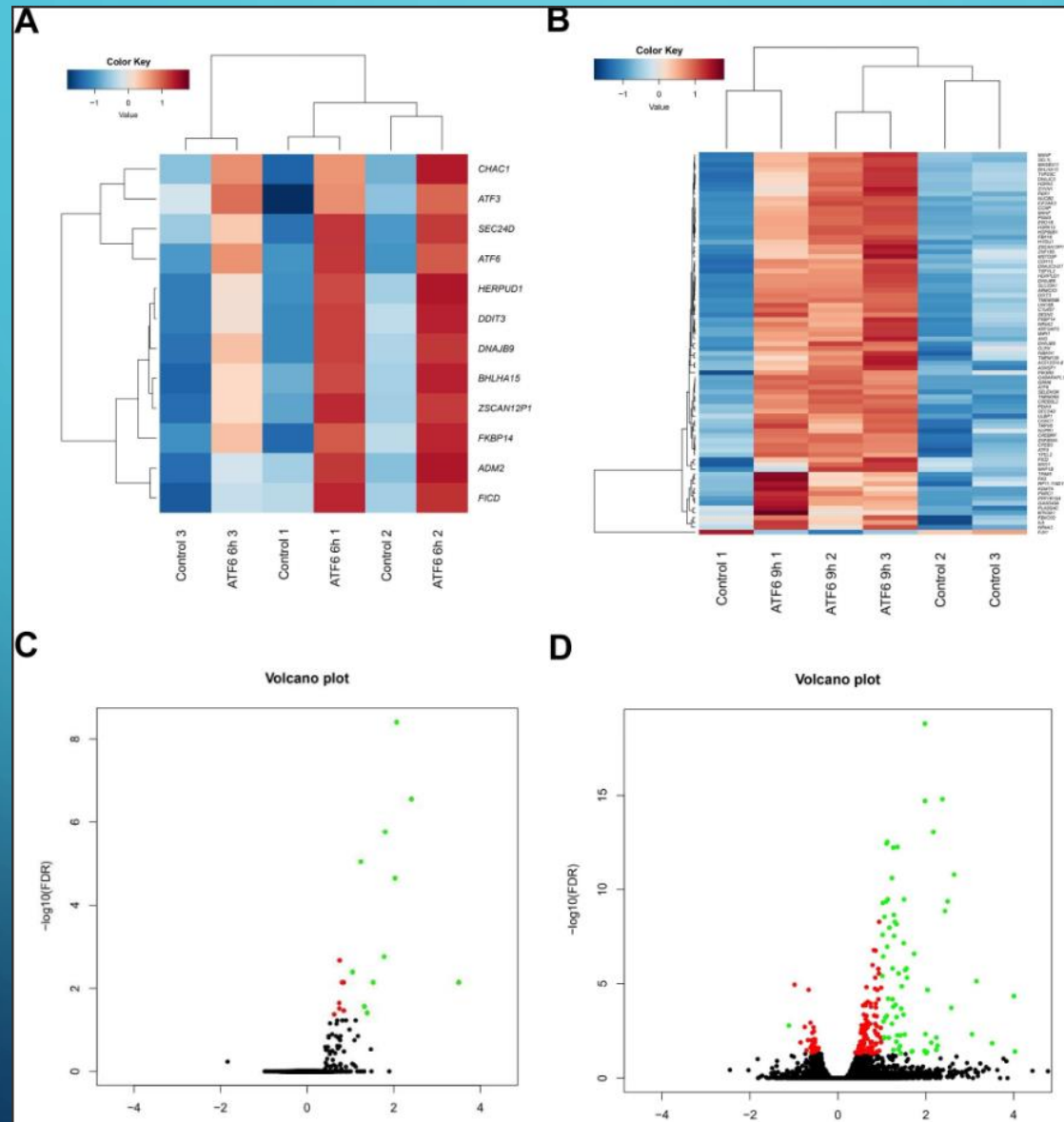
**Series** **GSE270286**

[Query DataSets for GSE270286](#)

|                 |                                                                                                                                                                                                                                                                                                                                                 |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Status          | Public on Nov 06, 2024                                                                                                                                                                                                                                                                                                                          |
| Title           | Transcriptomic analysis of cellular senescence induced by ectopic expression of ATF6a in human breast cancer cells                                                                                                                                                                                                                              |
| Organism        | <a href="#">Homo sapiens</a>                                                                                                                                                                                                                                                                                                                    |
| Experiment type | Expression profiling by high throughput sequencing                                                                                                                                                                                                                                                                                              |
| Summary         | To investigate the molecular mechanism that triggers endoplasmic reticulum (ER) stress-induced cellular senescence in breast cancer cells, we established MCF7 cell lines exposed to ATF6a ectopic expression. We then performed gene expression profiling analysis using data obtained from RNA-seq of 9 different cells at three time points. |
| Overall design  | Comparative gene expression profiling analysis of RNA-seq data for MCF-7 cells from three distinct time points (baseline, 6 hours, and 9 hours after ATF6a ectopic expression), using Illumina HiSeq X10 and NovaSeq 6000 platforms with paired-end 100 base sequencing.                                                                        |
| Contributor(s)  | <a href="#">Kim J</a> , <a href="#">Bae S</a> , <a href="#">Moon Y</a> , <a href="#">Kim E</a> , <a href="#">Kim Y</a> , <a href="#">Park Y</a> , <a href="#">Han M</a> , <a href="#">Sohn J</a>                                                                                                                                                |
| Citation(s)     | Kim JW, Bae SH, Moon Y, Kim EK et al. Transcriptomic analysis of cellular senescence induced by ectopic expression of ATF6a in human breast cancer cells. <i>PLoS One</i> 2024;19(10):e0309749. PMID: <a href="#">39466820</a>                                                                                                                  |

Series accession  
number

# LET'S TRY TO REPRODUCE THEIR RESULTS



# SEQUENCE READ ARCHIVE (SRA) RUN SELECTOR TO SEE SEQUENCING FILES

| Supplementary file | Size    | Download                                        | File type/resource |
|--------------------|---------|-------------------------------------------------|--------------------|
| GSE270286_RAW.tar  | 48.1 Mb | <a href="#">(http)</a> <a href="#">(custom)</a> | TAR (of TXT)       |

[SRA Run Selector](#) 

*Raw data are available in SRA*

# ACCESSION IDS

| Found 9 Items                                                |             |              |        |         |             |                       |              |                      |             |                                      |
|--------------------------------------------------------------|-------------|--------------|--------|---------|-------------|-----------------------|--------------|----------------------|-------------|--------------------------------------|
| <input checked="" type="checkbox"/> <input type="checkbox"/> | Run         | BioSample    | Bases  | Bytes   | Experiment  | Instrument            | Library Name | create_date          | Sample Name | treatment                            |
| <input type="checkbox"/> 1                                   | SRR29476514 | SAMN41934402 | 6.33 G | 1.84 Gb | SRX24988435 | Illumina NovaSeq 6000 | GSM8338904   | 2024-06-20 14:05:00Z | GSM8338904  | ATF6alpha overexpression for 9 hours |
| <input type="checkbox"/> 2                                   | SRR29476515 | SAMN41934403 | 6.32 G | 1.83 Gb | SRX24988434 | HiSeq X Ten           | GSM8338903   | 2024-06-20 14:07:00Z | GSM8338903  | ATF6alpha overexpression for 9 hours |
| <input type="checkbox"/> 3                                   | SRR29476516 | SAMN41934404 | 5.44 G | 1.58 Gb | SRX24988433 | HiSeq X Ten           | GSM8338902   | 2024-06-20 14:05:00Z | GSM8338902  | ATF6alpha overexpression for 9 hours |
| <input type="checkbox"/> 4                                   | SRR29476517 | SAMN41934405 | 6.32 G | 1.85 Gb | SRX24988432 | Illumina NovaSeq 6000 | GSM8338901   | 2024-06-20 14:05:00Z | GSM8338901  | ATF6alpha overexpression for 6 hours |
| <input type="checkbox"/> 5                                   | SRR29476518 | SAMN41934406 | 5.95 G | 1.74 Gb | SRX24988431 | HiSeq X Ten           | GSM8338900   | 2024-06-20 14:04:00Z | GSM8338900  | ATF6alpha overexpression for 6 hours |
| <input type="checkbox"/> 6                                   | SRR29476519 | SAMN41934407 | 4.94 G | 1.46 Gb | SRX24988430 | HiSeq X Ten           | GSM8338899   | 2024-06-20 14:04:00Z | GSM8338899  | ATF6alpha overexpression for 6 hours |
| <input type="checkbox"/> 7                                   | SRR29476520 | SAMN41934408 | 6.37 G | 1.83 Gb | SRX24988429 | Illumina NovaSeq 6000 | GSM8338898   | 2024-06-20 14:07:00Z | GSM8338898  | non-ATF6alpha overexpression         |
| <input type="checkbox"/> 8                                   | SRR29476521 | SAMN41934409 | 6.09 G | 1.78 Gb | SRX24988428 | HiSeq X Ten           | GSM8338897   | 2024-06-20 14:06:00Z | GSM8338897  | non-ATF6alpha overexpression         |
| <input type="checkbox"/> 9                                   | SRR29476522 | SAMN41934410 | 4.82 G | 1.41 Gb | SRX24988427 | HiSeq X Ten           | GSM8338896   | 2024-06-20 14:06:00Z | GSM8338896  | non-ATF6alpha overexpression         |

Individual  
sequencing runs

Important  
information  
regarding samples

# SRA TOOLKIT



[Systems](#) [Applications](#) [Reference Data](#) [Storage](#) [User Guides](#) [Training](#) [User Dashboard](#) [How To](#) [About](#)

## SRA-Toolkit

The [NCBI SRA SDK](#) generates loading and dumping tools with their respective libraries for building new and accessing existing runs.

## Documentation

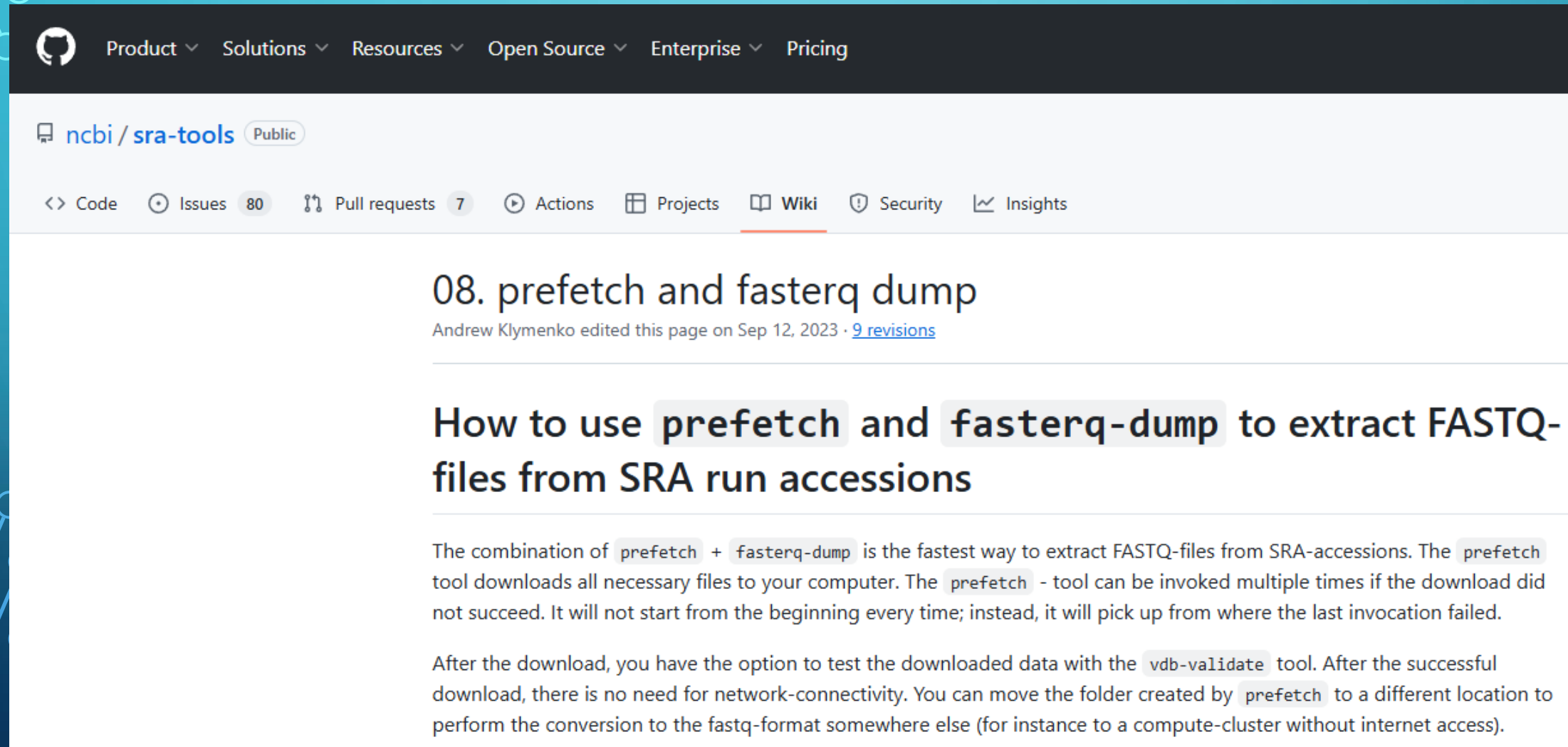
- [NCBI SRA Download Guide](#)
- [SRA Toolkit documentation](#)
- [SRA File Formats Guide](#)
- **Command line help:** Type the command followed by '-h'
- [fasterq-dump guide](#)

## Quick Links

[SRA Source Repositories](#)  
[Estimating space requirements](#)  
[Downloading from NCBI](#)  
[Batch job on Biowulf](#)  
[Using Swarm](#)  
[Interactive jobs](#)  
[dbGaP downloads](#)  
[Executables](#)  
[Documentation](#)  
[Configuring SRA-Toolkit](#)  
[Troubleshooting](#)

## Important Notes

# PREFETCH AND FASTERQ-DUMP



The screenshot shows the GitHub interface for the repository `ncbi/sra-tools`. The top navigation bar includes links for Product, Solutions, Resources, Open Source, Enterprise, and Pricing. Below this, the repository name `ncbi/sra-tools` is displayed with a 'Public' label. A secondary navigation bar contains icons and labels for Code, Issues (80), Pull requests (7), Actions, Projects, Wiki (which is the active tab), Security, and Insights.

## 08. prefetch and fasterq dump

Andrew Klymenko edited this page on Sep 12, 2023 · [9 revisions](#)

### How to use `prefetch` and `fasterq-dump` to extract FASTQ-files from SRA run accessions

The combination of `prefetch` + `fasterq-dump` is the fastest way to extract FASTQ-files from SRA-accessions. The `prefetch` tool downloads all necessary files to your computer. The `prefetch` - tool can be invoked multiple times if the download did not succeed. It will not start from the beginning every time; instead, it will pick up from where the last invocation failed.

After the download, you have the option to test the downloaded data with the `vdb-validate` tool. After the successful download, there is no need for network-connectivity. You can move the folder created by `prefetch` to a different location to perform the conversion to the fastq-format somewhere else (for instance to a compute-cluster without internet access).

# USE A SCRIPT TO DOWNLOAD SEQUENCING DATA

Host: localhost

GNU nano 2.9.8

```
#!/bin/bash
```

Specifies 'bash' language of script

```
#SBATCH -J GSE270286
```

```
#SBATCH -o GSE270286_sra.out
```

```
#SBATCH -N 1 -n 32
```

- Job name
- Log file produced
- Requesting one whole node (32 cores)

```
module load sratoolkit
```

Loads the required sratoolkit module

```
while read p; do
```

```
    prefetch "$p"
```

```
    fasterq-dump -e 32 --outdir ./"$p"/ "$p"
```

```
    pigz -p 32 ./"$p"/*.fastq
```

```
    rm ./"$p"/*.sra
```

```
done < GSE270286.txt
```

# USE A SCRIPT TO DOWNLOAD SEQUENCING DATA

Host: localhost

GNU nano 2.9.8

```
#!/bin/bash
```

```
#SBATCH -J GSE270286
```

```
#SBATCH -o GSE270286_sra.out
```

```
#SBATCH -N 1 -n 32
```

```
module load sratoolkit
```

```
while read p; do
```

```
    prefetch "$p"
```

```
    fasterq-dump -e 32 --outdir ./"$p"/ "$p"
```

```
    pigz -p 32 ./"$p"/*.fastq
```

```
    rm ./"$p"/*.sra
```

```
done < GSE270286.txt
```

'while loop' that reads each line of GSE270286.txt and replaces the variable 'p' with the accession run (SRR...)

Input for 'while loop'

- Prefetch each accession run ("p" variable)
- download each run (in parallel (32 cores))
- Unzip each run (in parallel)
- Remove unnecessary files

# COPY OVER SCRIPT AND SRR LIST TO YOUR DIRECTORY

Script:

```
[databiom@tsu-hpc01 linux]$ cp /home/databiom/linux/sratoolkit_prefetch_dump_batch.sh .
```

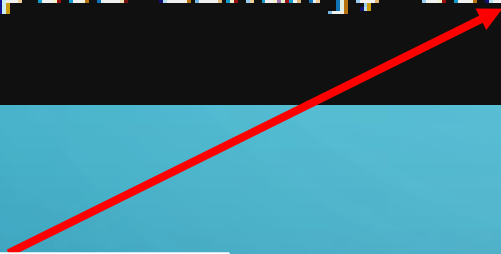
SRR List:

```
[databiom@tsu-hpc01 linux]$ cp /home/databiom/linux/GSE270286.txt .
```

# SUBMIT JOB TO EXECUTE SCRIPT

```
[databiom@tsu-hpc01 linux]$ sbatch sratoolkit_prefetch_dump_batch.sh
```

Command to  
submit job to job  
scheduler



# USE THE 'QUEUE' COMMAND TO CHECK THE JOB QUEUE FOR YOUR JOB

```
[databion@tsu-hpc01 linux]$ squeue
```

| JOBID | PARTITION | NAME     | USER | ST | TIME    | NODES | ODELIST(Reason) |
|-------|-----------|----------|------|----|---------|-------|-----------------|
| 4220  | defq      | gpu_md   |      | R  | 3:13:04 | 1     | gpu003          |
| 4221  | defq      | gpu_md   |      | R  | 3:12:23 | 1     | gpu004          |
| 4222  | defq      | gpu_md   |      | R  | 3:11:41 | 1     | gpu005          |
| 4223  | defq      | gpu_md   |      | R  | 3:10:50 | 1     | gpu006          |
| 4224  | defq      | gpu_md   |      | R  | 3:10:14 | 1     | gpu007          |
| 4225  | defq      | gpu_md   |      | R  | 3:09:23 | 1     | gpu008          |
| 4226  | defq      | gpu_md   |      | R  | 3:08:24 | 1     | gpu009          |
| 4227  | defq      | gpu_md   |      | R  | 3:07:26 | 1     | gpu010          |
| 4229  | defq      | ::app=Li |      | R  | 50:56   | 1     | gpu001          |
| 4230  | defq      | ::app=Li |      | R  | 48:48   | 1     | gpu011          |
| 4250  | defq      | gpu_md   |      | R  | 15:02   | 1     | gpu012          |
| 4251  | defq      | gpu_md   |      | R  | 14:56   | 1     | gpu013          |
| 4252  | defq      | gpu_md   |      | R  | 14:50   | 1     | gpu014          |
| 4253  | defq      | gpu_md   |      | R  | 14:46   | 1     | gpu015          |

```
[databion@tsu-hpc01 linux]$
```



# GOOD JOB!

LINUX & HPC